Modelling and Forecasting Singapore's Total Fertility Rate and Total Live Births Using ARIMA and SARIMA Models

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GitHub repository:
`https://github.com/Bubble128/MATH3131-TimeSeries-Assignment1`

Abstract

This report analyses annual Singapore demographic time series data, focusing on Total Fertility Rate (TFR) and Total Live Births (TLB) from 1960 to 2024. The main aim is to identify suitable time series models for describing the historical behaviour of these series and forecasting the held-out period from 2013 to 2024. The models are fitted using the training period from 1960 to 2012 and evaluated against the testing period from 2013 to 2024. To stabilise variation and improve model identification, both series are first analysed on the log scale. Baseline ARIMA models are considered, but residual diagnostics indicate that these simple models leave remaining autocorrelation. Since the ACF and PACF of the differenced log series show evidence of period-12 dependence, SARIMA models with $D = 1$ and seasonal period 12 are then examined. The ACF and PACF after both ordinary and seasonal differencing are used to guide the final SARIMA orders. Model adequacy is assessed using AIC, BIC, Ljung–Box residual tests, and out-of-sample forecast accuracy. The preferred models are ARIMA$(1, 1, 0)(0, 1, 1)_{12}$ for log(TLB) and ARIMA$(4, 1, 0)(0, 1, 1)_{12}$ for log(TFR), as these models produce acceptable residual diagnostics and comparatively strong forecast performance over the testing period.

1 Introduction

Singapore has experienced substantial demographic change since the 1960s, including a long-term decline in fertility and changes in the annual number of live births. These changes are important because fertility and birth trends are closely related to population ageing, labour supply, and long-term social planning. Recent official population statistics show that Singapore's resident total fertility rate (TFR) remained at 0.97 in 2024, unchanged from 2023, while citizen births in 2024 were still lower on average than in the preceding five-year period (National Population and Talent Division, 2025). The same official report also notes that Singaporean families have become smaller over time, with more ever-married females having no children or only one child, and fewer having three or more children (National Population and Talent Division, 2025). These patterns make fertility and birth data important subjects for statistical modelling.

This report analyses two annual Singapore demographic time series: Total Fertility Rate (TFR) and Total Live Births (TLB). The data are obtained from the official *Births And Fertility Rates, Annual* dataset published on data.gov.sg by the Singapore Department of Statistics (Singapore Department of Statistics, 2023). The downloaded CSV file, `BirthsAndFertilityRatesAnnual.csv`, contains annual demographic series including Total Fertility Rate and Total Live Births. TFR measures the average number of live births that a female would have during her reproductive years under the prevailing age-specific fertility rates, while TLB records the annual number of live births. Together, these two series provide complementary views of Singapore's demographic change: TFR describes fertility behaviour relative to the female population of reproductive age, whereas TLB reflects the realised number of births in each year.

Time series models are appropriate for this setting because both fertility rates and live births evolve over time and may contain trend, autocorrelation, and periodic structure. Previous demographic research has used ARIMA models to study birth and pregnancy patterns. For example, Yamamoto et al. applied ARIMA modelling to Japanese monthly birth and pregnancy data during the COVID-19 pandemic, following the identification, estimation, and forecasting stages of ARIMA analysis (Yamamoto et al., 2025). Their study demonstrates that ARIMA-type methods can be used to model demographic birth series and compare observed outcomes against forecasts. This provides methodological motivation for applying ARIMA and SARIMA-type models to Singapore's fertility and birth data.

The main research question of this report is: how can ARIMA and SARIMA-type models be used to model and forecast Singapore's Total Fertility Rate and Total Live Births, and does the inclusion of a period-12 component improve the modelling of these annual demographic series? To answer this question, the series are modelled using data from 1960 to 2012 and evaluated against the held-out period from 2013 to 2024. Baseline ARIMA models are first considered, followed by SARIMA-type models motivated by the autocorrelation and partial autocorrelation patterns after ordinary and seasonal differencing. Model adequacy is assessed using information criteria, Ljung–Box residual tests, and out-of-sample forecast accuracy.

The remainder of the report is structured as follows. Section 2 describes the data, transformations, and train–test split. Section 3 outlines the ARIMA and SARIMA modelling approach. Section 4 presents the model comparison and forecasting results. Section 5 discusses the interpretation and limitations of the selected models. Section 6 concludes.

2 Data and Exploratory Analysis

The data consist of annual observations of Singapore's Total Fertility Rate (TFR) and Total Live Births (TLB) from 1960 to 2024. The observations from 1960 to 2012 are used as the training set, while the observations from 2013 to 2024 are held out as the testing set for forecast evaluation. This train–test split follows the aim of assessing whether models fitted to the historical period can forecast the more recent period.

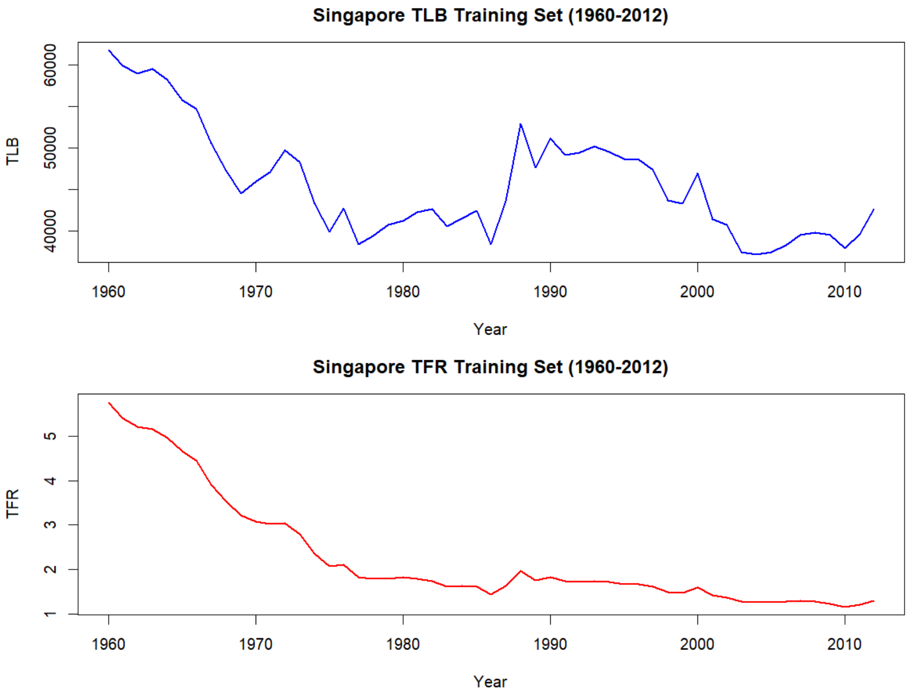

Figure 1: Original-scale training series of Singapore Total Live Births and Total Fertility Rate, 1960–2012.

Figure 1 shows that both series display clear long-term changes over the training period. TFR declined sharply from the 1960s and then remained at a much lower level in later decades. TLB also changed substantially over time, with a decline from the 1960s to the mid-1970s, a temporary rise around the late 1980s, and further variation afterwards. These visible trends suggest that the raw series are not stationary and should not be modelled directly using stationary ARMA-type models.

To stabilise the scale of the data and reduce the effect of changing variability, both series are analysed on the log scale. The log transformation is also useful because forecast errors on the log scale can be interpreted approximately in relative terms.

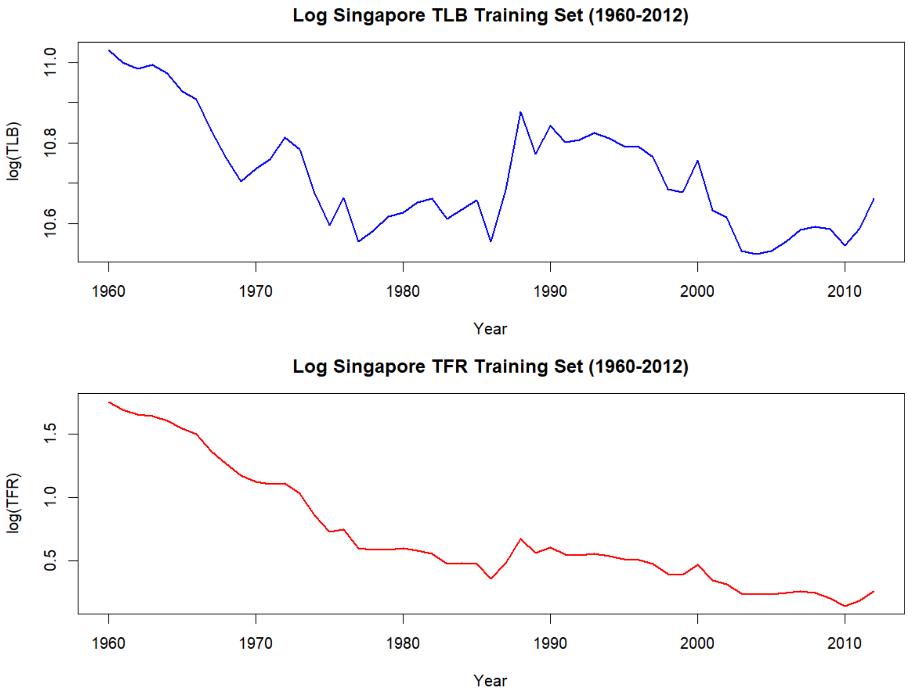

Figure 2: Log-transformed training series of Singapore Total Live Births and Total Fertility Rate, 1960–2012.

Figure 2 shows that the log transformation changes the scale of the series but does not remove the main downward trend. Therefore, ordinary first-order differencing is applied to the log-transformed series. The first-differenced log series are shown in Figure 3.

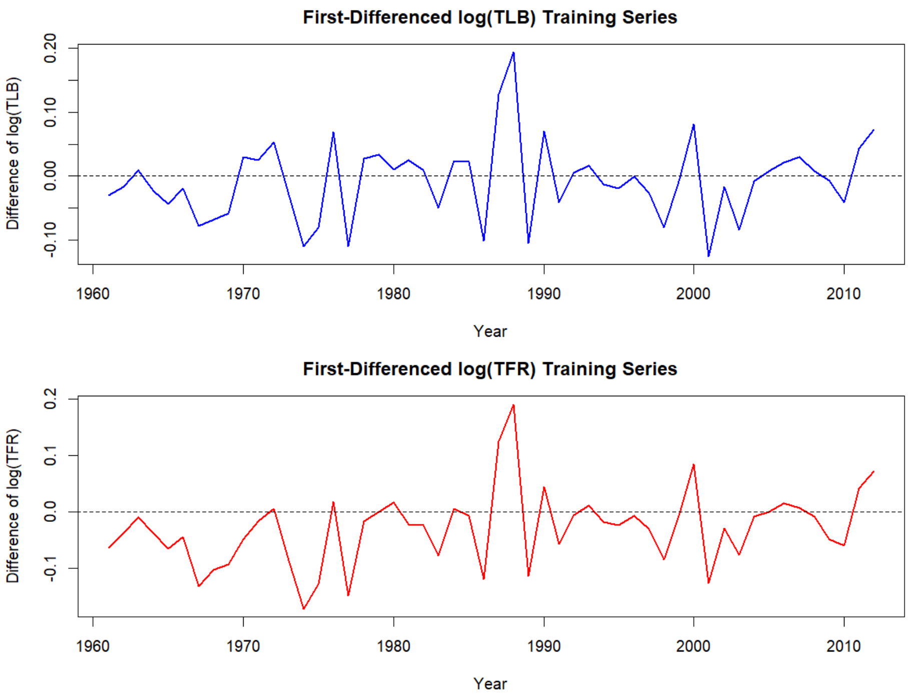

Figure 3: First-differenced log-transformed TLB and TFR training series.

After first differencing, both transformed series fluctuate around a roughly constant level, suggesting that $d = 1$ is a reasonable starting point. However, several large movements remain, particularly around the late 1980s and early 2000s, so the autocorrelation structure is examined more closely using the ACF and PACF.

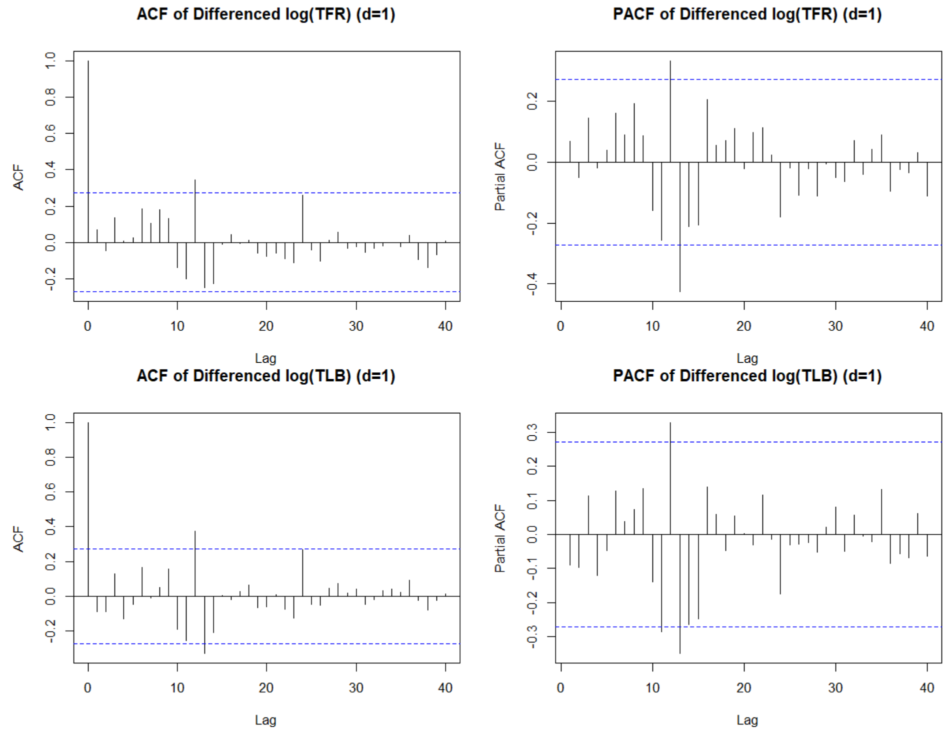

Figure 4: ACF and PACF of the first-differenced log-transformed TFR and TLB series, using lags up to 40.

Figure 4 shows that the first-differenced log series still contain some dependence at longer lags, especially around lag 12 and nearby lags. Since the data are annual, this should not be interpreted as ordinary within-year seasonality. Instead, it suggests the possibility of a longer period-12 demographic cycle. This motivates the use of SARIMA-style models with seasonal period $s = 12$ as candidate models.

To identify SARIMA orders more directly, both ordinary first differencing and period-12 seasonal differencing are applied to the log-transformed training series. This corresponds to using $d = 1$ and $D = 1$ when considering models of the form ARIMA$(p, 1, q)(P, 1, Q)_{12}$.

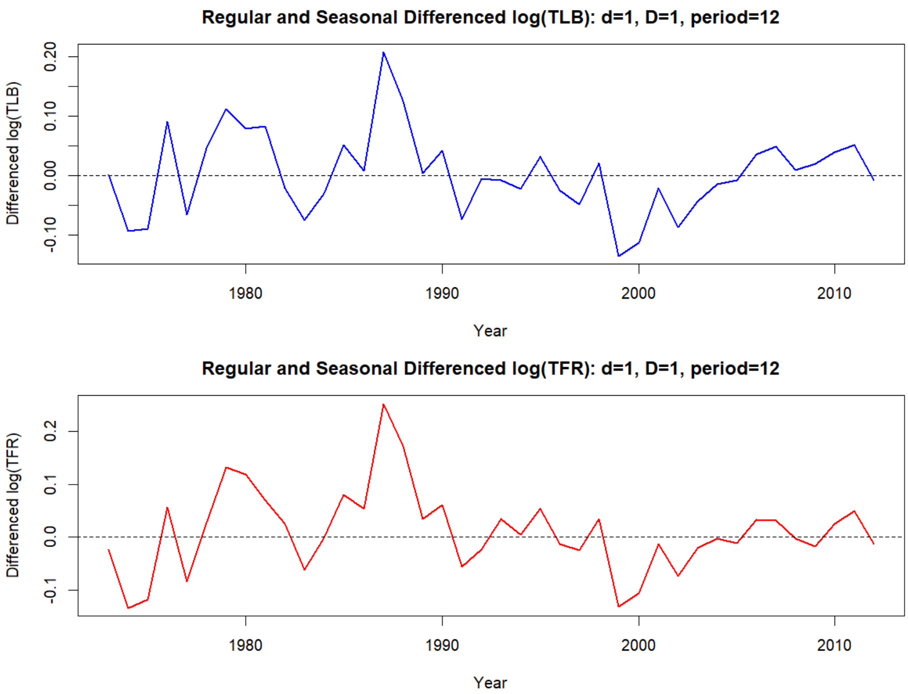

Figure 5: Log-transformed TLB and TFR after ordinary first differencing and period-12 seasonal differencing.

Figure 5 suggests that the additional period-12 differencing removes much of the remaining long-cycle structure. The corresponding ACF and PACF plots are shown in Figure 6. These plots are used to guide the SARIMA candidate orders.

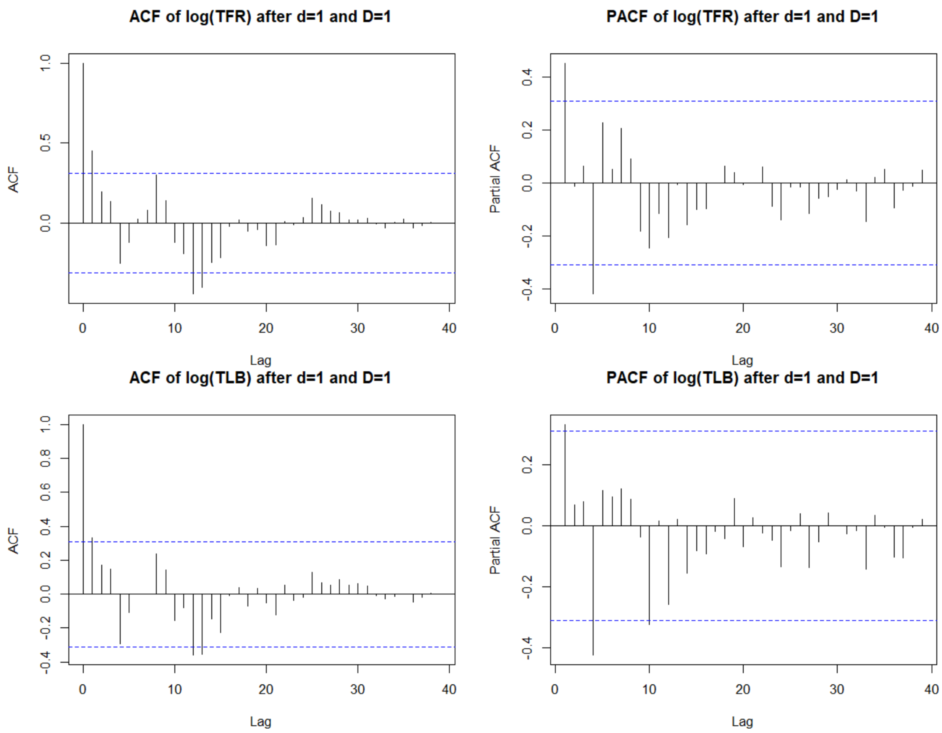

Figure 6: ACF and PACF of log-transformed TFR and TLB after ordinary first differencing and period-12 seasonal differencing.

For TLB, the PACF after both differences shows some short-lag structure and a notable feature around lag 4, while the ACF shows remaining structure around lag 12. This motivates both low-order SARIMA candidates and lag-4 SARIMA candidates. The low-order candidates are included as simpler models, while the lag-4 candidates reflect the visible PACF feature.

For TFR, the PACF also suggests possible non-seasonal autoregressive structure around lag 4, while the ACF suggests short-lag moving-average behaviour and remaining period-12 dependence. This motivates SARIMA candidates with a period-12 moving-average component and alternative non-seasonal ARMA structures.

3 Methods

3.1 ARIMA Baseline Models

An ARIMA(p, d, q) model combines autoregressive terms of order p, ordinary differencing of order d, and moving-average terms of order q. Since the original and log-transformed series show clear non-stationary behaviour, first-order differencing is used for the baseline ARIMA models. These baseline models provide simple non-seasonal benchmarks before considering the period-12 SARIMA-style models.

For TLB, the baseline ARIMA models considered are

$$\text{ARIMA}(0,1,0), \quad \text{ARIMA}(0,1,1), \quad \text{ARIMA}(1,1,0).$$

These models represent a random-walk-type benchmark and two simple low-order ARIMA alternatives.

For TFR, the baseline ARIMA models considered are

$$\text{ARIMA}(1,1,0), \quad \text{ARIMA}(0,1,1), \quad \text{ARIMA}(1,1,1).$$

These models are included because the ACF and PACF of the first-differenced log series suggest short-lag dependence. In addition, `auto.arima` is used as a non-seasonal automatic benchmark for each series. The automatic models are not treated as final choices by themselves, but are included to compare the manually identified models with a standard automatic selection procedure.

3.2 SARIMA-Style Period-12 Models

Because the ACF and PACF of the first-differenced log series show possible dependence around lag 12, SARIMA-style models with period $s = 12$ are also considered. Since the observations are annual, this period is not interpreted as ordinary within-year seasonality.

Instead, it is treated as a possible 12-year demographic cycle.

A SARIMA model can be written as

$$\mathrm{ARIMA}(p, d, q)(P, D, Q)_s,$$

where p, d, q describe the non-seasonal ARIMA component, P, D, Q describe the period-s component, and s is the seasonal or cyclic period.

After applying ordinary first differencing and period-12 differencing to the log-transformed training series, the ACF and PACF are used to identify candidate values of p, q, P, and Q. This gives $d = 1$, $D = 1$, and $s = 12$ for the SARIMA-style candidates.

For TLB, the PACF after both differencing steps shows a notable spike around lag 4, while the ACF shows remaining structure around lag 12. Therefore, both low-order SARIMA-style candidates and lag-4 candidates are considered:

$$\begin{aligned}
&\mathrm{ARIMA}(1, 1, 0)(0, 1, 1)_{12},\\
&\mathrm{ARIMA}(0, 1, 1)(0, 1, 1)_{12},\\
&\mathrm{ARIMA}(4, 1, 0)(0, 1, 1)_{12},\\
&\mathrm{ARIMA}(4, 1, 1)(0, 1, 1)_{12}.
\end{aligned}$$

The low-order SARIMA candidates are included as simpler period-12 models, while the models with $p = 4$ are motivated by the non-seasonal PACF pattern. In all four candidates, $P = 0$ and $Q = 1$ are motivated by the stronger period-12 structure in the ACF than in the PACF.

For TFR, the PACF suggests possible non-seasonal autoregressive structure around lag 4, while the ACF suggests short-lag moving-average behaviour and remaining period-12 dependence. Therefore, the SARIMA-style candidates considered for TFR are

$$\begin{aligned}
&\mathrm{ARIMA}(1, 1, 1)(0, 1, 1)_{12},\\
&\mathrm{ARIMA}(1, 1, 2)(0, 1, 1)_{12},\\
&\mathrm{ARIMA}(4, 1, 0)(0, 1, 1)_{12},\\
&\mathrm{ARIMA}(4, 1, 1)(0, 1, 1)_{12},\\
&\mathrm{ARIMA}(4, 1, 2)(0, 1, 1)_{12}.
\end{aligned}$$

The lower-order candidates are included as simpler alternatives, while the models with $p = 4$ are motivated by the lag-4 behaviour in the PACF after both ordinary and period-12 differencing. The models with $q = 1$ or $q = 2$ are included because the ACF suggests short-lag moving-average structure. The model with $q = 0$ is also retained as a simpler comparison model to assess whether the additional moving-average terms improve residual diagnostics and forecast accuracy.

3.3 Model Evaluation

Models are evaluated using residual diagnostics, information criteria, and out-of-sample forecast accuracy. Residual diagnostics are used to assess whether the fitted model has removed the main autocorrelation structure from the data. In particular, the Ljung–Box test is used to test whether residual autocorrelations remain jointly significant. A model with a Ljung–Box p-value below 0.05 is treated as inadequate because this suggests remaining autocorrelation in the residuals.

AIC and BIC are used to compare in-sample model fit while penalising model complexity. However, information criteria are not used alone to select the final model, because a model with a low AIC or BIC may still leave autocorrelated residuals. Therefore, the final model selection gives priority to residual adequacy and then compares forecast accuracy among the residual-valid models.

Out-of-sample forecast accuracy is assessed using the testing period from 2013 to 2024. The main forecast error measures are the sum of squared errors (SSE), root mean squared error (RMSE), and mean absolute error (MAE). These are defined as

$$\mathrm{SSE} = \sum_{t=1}^{h} (y_t - \hat{y}_t)^2,$$

$$\mathrm{RMSE} = \sqrt{\frac{1}{h} \sum_{t=1}^{h} (y_t - \hat{y}_t)^2},$$

and

$$\mathrm{MAE} = \frac{1}{h} \sum_{t=1}^{h} |y_t - \hat{y}_t|,$$

where y_t denotes the observed testing value, $\hat{y}_t$ denotes the corresponding forecast, and h is the forecast horizon. Since the models are fitted on the log scale, these error measures are also calculated on the log scale.

4 Results

This section compares the candidate models fitted to the log-transformed TFR and TLB series. The models are evaluated using AIC, BIC, Ljung–Box residual test p-values, and out-of-sample forecast accuracy over the 2013–2024 testing period. Since the models are fitted on the log scale, the RMSE and MAE values reported in the tables are also calculated on the log scale.

4.1 TFR Model Comparison

Table 1: Candidate model comparison for log(TFR).

Model	AIC	BIC	Ljung–Box p	Test RMSE	Test MAE
ARIMA$(1,1,0)$	-124.526	-120.624	0.0026	0.1790	0.1591
ARIMA$(0,1,1)$	-124.233	-120.330	0.0045	0.1720	0.1515
ARIMA$(1,1,1)$	-127.787	-121.933	0.0048	0.1375	0.1172
ARIMA$(1,1,1)(0,1,1)_{12}$	-104.922	-98.166	0.5175	0.1349	0.1268
ARIMA$(1,1,2)(0,1,1)_{12}$	-103.433	-94.988	0.5201	0.1365	0.1285
ARIMA$(4,1,0)(0,1,1)_{12}$	-102.761	-92.627	0.7649	0.1231	0.1125
ARIMA$(4,1,1)(0,1,1)_{12}$	-104.312	-92.490	0.8387	0.1274	0.1168
ARIMA$(4,1,2)(0,1,1)_{12}$	-102.318	-88.807	0.7964	0.1269	0.1162
`auto.arima`	-125.644	-121.781	0.0048	0.1084	0.0910

Table 1 shows that the simple non-seasonal ARIMA models do not provide adequate residual diagnostics for TFR. Although ARIMA$(1,1,1)$ improves the forecast accuracy compared with the simpler ARIMA baselines, its Ljung–Box p-value remains below 0.05, indicating remaining autocorrelation in the residuals. The `auto.arima` benchmark gives the lowest testing RMSE and MAE, but it also fails the Ljung–Box residual test, so it is not considered adequate as a final model.

The SARIMA-style period-12 models perform better in terms of residual adequacy. All five SARIMA candidates have Ljung–Box p-values above 0.05, suggesting that the residual autocorrelation has been substantially reduced. Among the residual-valid SARIMA models, ARIMA$(4,1,0)(0,1,1)_{12}$ gives the lowest testing RMSE and MAE. Therefore, ARIMA$(4,1,0)(0,1,1)_{12}$ is selected as the preferred model for log(TFR). The close performance of ARIMA$(4,1,1)(0,1,1)_{12}$ and ARIMA$(4,1,2)(0,1,1)_{12}$ suggests that the fourth-order autoregressive component is important, while additional moving-average terms do not improve forecast accuracy for this series.

4.2 TLB Model Comparison

Table 2: Candidate model comparison for log(TLB).

Model	AIC	BIC	Ljung–Box p	Test RMSE	Test MAE
ARIMA$(0, 1, 0)$	-140.938	-138.987	0.0047	0.1267	0.1023
ARIMA$(0, 1, 1)$	-139.267	-135.365	0.0077	0.1215	0.0958
ARIMA$(1, 1, 0)$	-139.225	-135.323	0.0067	0.1226	0.0972
ARIMA$(1, 1, 0)(0, 1, 1)_{12}$	-109.214	-104.147	0.0817	0.0721	0.0608
ARIMA$(0, 1, 1)(0, 1, 1)_{12}$	-109.182	-104.116	0.0667	0.0724	0.0614
ARIMA$(4, 1, 0)(0, 1, 1)_{12}$	-108.319	-98.186	0.2050	0.0822	0.0711
ARIMA$(4, 1, 1)(0, 1, 1)_{12}$	-109.728	-97.906	0.4846	0.0844	0.0747
`auto.arima`	-140.938	-138.987	0.0047	0.1267	0.1023

Table 2 shows a similar pattern for TLB. The simple ARIMA baseline models have comparatively low AIC and BIC values, but their Ljung–Box p-values are all below 0.05. This indicates that these models leave significant residual autocorrelation and are therefore not adequate final models. The `auto.arima` benchmark selects a model equivalent to the simple random-walk-type ARIMA baseline, and it also fails the residual diagnostic.

The SARIMA-style models provide much stronger residual diagnostics. All four SARIMA candidates have Ljung–Box p-values above 0.05, indicating no strong evidence of remaining residual autocorrelation. Among these residual-valid models, ARIMA$(1, 1, 0)(0, 1, 1)_{12}$ gives the lowest testing RMSE and MAE, with ARIMA$(0, 1, 1)(0, 1, 1)_{12}$ performing very similarly. Therefore, ARIMA$(1, 1, 0)(0, 1, 1)_{12}$ is selected as the preferred model for log(TLB). The lag-4 SARIMA models also pass the residual diagnostic, but their forecast accuracy is weaker than the simpler low-order SARIMA models.

4.3 Forecasts

The selected models are used to forecast the held-out period from 2013 to 2024. For TFR, the preferred model is ARIMA$(4, 1, 0)(0, 1, 1)_{12}$ fitted to log(TFR). For TLB, the preferred model is ARIMA$(1, 1, 0)(0, 1, 1)_{12}$ fitted to log(TLB). Forecasts are transformed back to the original scale for visual comparison with the observed testing data.

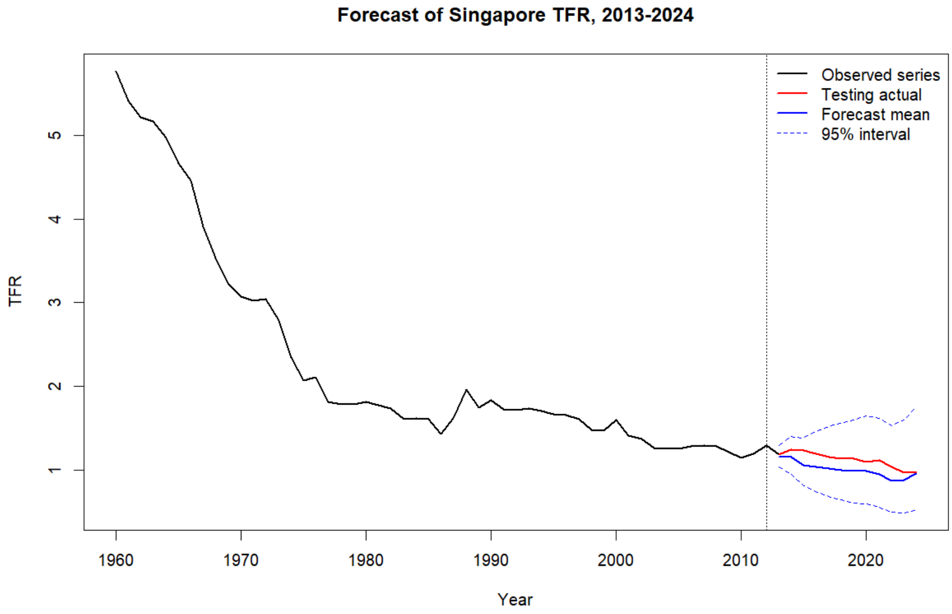

Figure 7: Forecast comparison for TFR over the testing period 2013–2024.

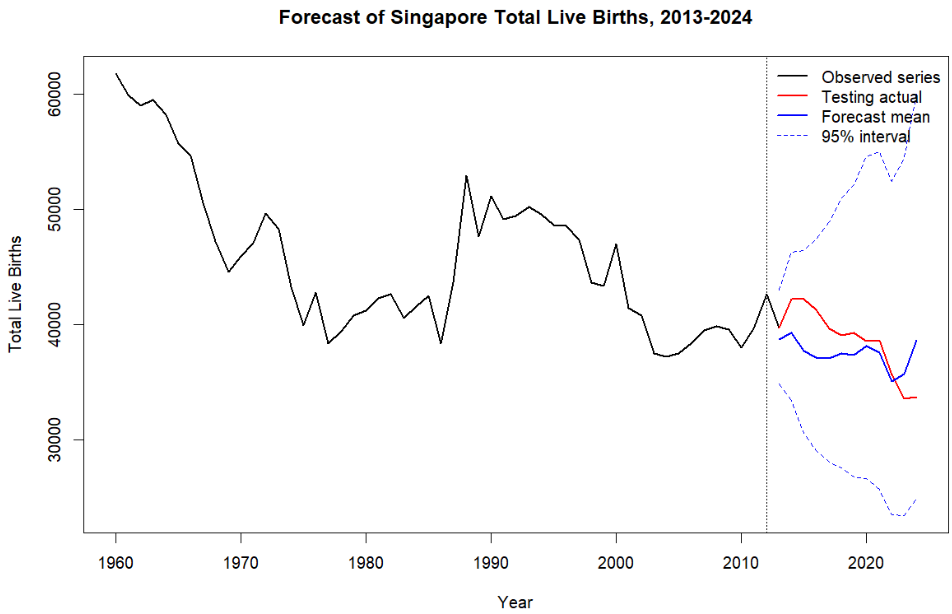

Figure 8: Forecast comparison for TLB over the testing period 2013–2024.

Figures 7 and 8 compare the forecasts from the selected models with the observed testing data after transforming the forecasts back to the original scale. The TFR forecast follows the broad low-fertility level in the testing period, although some year-to-year changes are not captured exactly. The TLB forecast is more conservative and does not fully reproduce the later decline and rebound in the observed testing values. Overall, the forecast plots support the numerical accuracy results: the selected SARIMA-style models provide useful forecasts, but they should be interpreted with caution because demographic series can be affected by external factors not included in the models.

4.4 Residual Diagnostics for Selected Models

The residual diagnostics for the selected final models are shown in Figures 9 and 10. These plots provide a visual check of whether the selected SARIMA-style models have removed the main autocorrelation structure from the data.

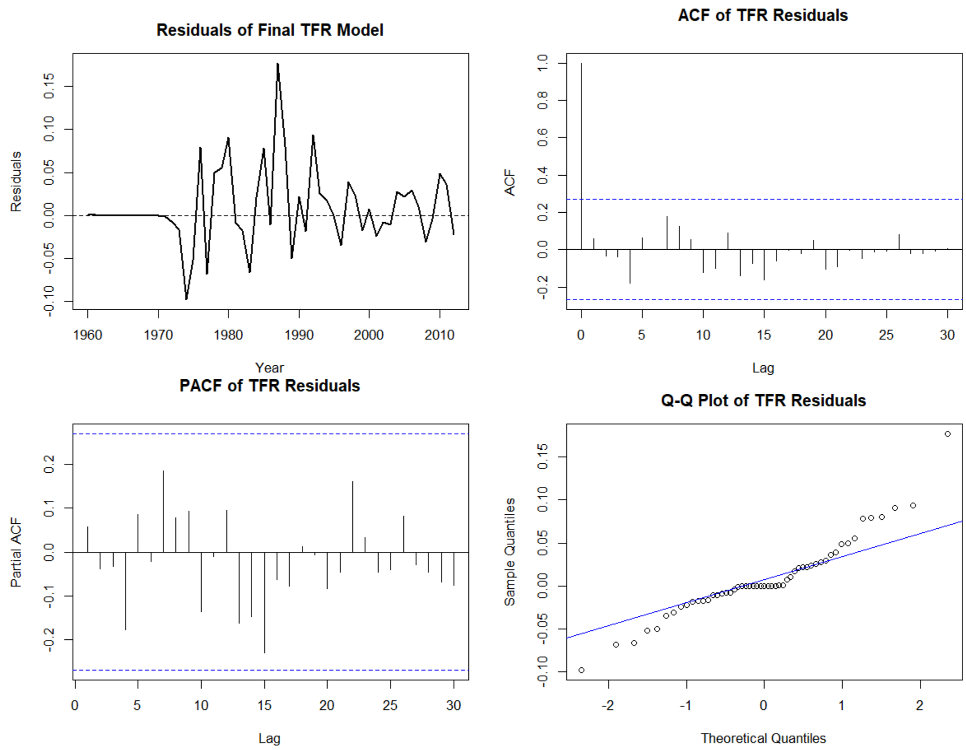

Figure 9: Residual diagnostics for the selected ARIMA$(4,1,0)(0,1,1)_{12}$ model fitted to log(TFR).

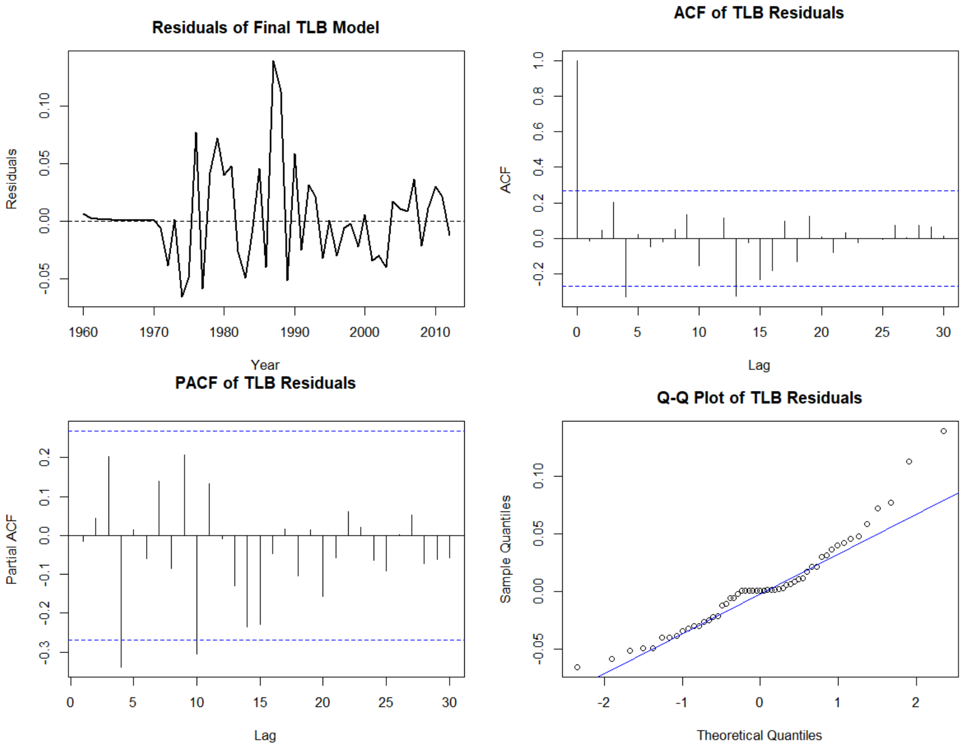

Figure 10: Residual diagnostics for the selected ARIMA$(1,1,0)(0,1,1)_{12}$ model fitted to log(TLB).

Together with the Ljung–Box p-values reported in Tables 1 and 2, these diagnostic plots support the adequacy of the selected models. The residual autocorrelation is substantially reduced compared with the baseline ARIMA models, although some individual fluctuations remain.

5 Discussion

The results show that the inclusion of a period-12 SARIMA-style component improves the modelling of both demographic series. For both TFR and TLB, the simple non-seasonal ARIMA benchmark models have relatively low AIC and BIC values, but their Ljung–Box p-values are below 0.05. This indicates that these baseline models still leave significant autocorrelation in the residuals. Therefore, although the baseline ARIMA models provide useful starting points, they are not adequate final models for these data.

For TFR, the preferred model is ARIMA$(4,1,0)(0,1,1)_{12}$ fitted to log(TFR). This model has an acceptable Ljung–Box p-value and the lowest RMSE and MAE among the SARIMA candidates. The result suggests that the period-12 component helps to remove remaining long-lag dependence that is not captured by the baseline ARIMA models. The fourth-order non-seasonal autoregressive term is also important, which is consistent with the PACF pattern after ordinary and period-12 differencing. Although `auto.arima` gives the lowest forecast error for TFR, it fails the residual diagnostic test, so it is not selected as the final model. This illustrates why forecast accuracy alone is not sufficient for model selection.

For TLB, the preferred model is ARIMA$(1, 1, 0)(0, 1, 1)_{12}$ fitted to log(TLB). This model has the lowest testing RMSE and MAE among the residual-valid SARIMA candidates. The lag-4 SARIMA models also pass the Ljung–Box test, but their out-of-sample forecast accuracy is weaker than the simpler low-order SARIMA models. This suggests that, for TLB, the period-12 moving-average component is important, but a more complex non-seasonal autoregressive structure does not improve forecasting performance.

The forecast plots also show different behaviour for the two series. For TFR, the selected SARIMA model captures the general low level of fertility over the testing period, although the forecast mean does not perfectly follow every year-to-year movement. For TLB, the forecast is more conservative and does not fully capture the later changes in the testing period. This is not surprising because live births are affected not only by fertility rates but also by the size and age structure of the population, migration, marriage patterns, economic conditions, and policy changes. These factors are not explicitly included in the univariate ARIMA and SARIMA models used in this report.

The period-12 component should also be interpreted carefully. Since the data are annual, this is not ordinary seasonality within a year. Instead, it may reflect a longer demographic cycle or repeated social pattern, possibly related to cohort effects or culturally meaningful timing patterns. However, the sample contains only 53 observations in the training period, so there are only a few complete 12-year cycles. This means that the evidence for a period-12 structure should be treated as suggestive rather than definitive.

There are several limitations to this analysis. First, the training sample is relatively small, which makes high-order models more vulnerable to overfitting. Second, the models are univariate and only use the past values of each series, so they cannot directly account for external demographic drivers such as fertility policy, economic conditions, housing costs, migration, or changes in marriage behaviour. Third, the forecasts are evaluated over only 12 testing observations, so small differences in RMSE and MAE should not be overinterpreted. Despite these limitations, the results provide evidence that SARIMA-style period-12 models give more adequate residual behaviour than simple ARIMA baselines for these Singapore demographic series.

6 Conclusion

This report modelled and forecast Singapore's Total Fertility Rate and Total Live Births using annual data from 1960 to 2024. The series were fitted on the log scale using the training period from 1960 to 2012 and evaluated against the held-out period from 2013 to 2024. The analysis first considered baseline ARIMA models and then extended the modelling to SARIMA-style models with a period-12 component.

The main finding is that the period-12 SARIMA-style models provide more adequate residual diagnostics than the simple ARIMA baselines. For log(TFR), the preferred model is ARIMA$(4, 1, 0)(0, 1, 1)_{12}$. For log(TLB), the preferred model is ARIMA$(1, 1, 0)(0, 1, 1)_{12}$. Both selected models pass the Ljung–Box residual diagnostic and achieve comparatively strong forecast accuracy among the residual-valid candidate models.

These results suggest that Singapore's fertility and birth series contain dependence that is not fully captured by simple non-seasonal ARIMA models. The inclusion of a period-12 component improves model adequacy, although this should be interpreted as a possible long-cycle demographic pattern rather than ordinary seasonality. Future work could extend the analysis by including explanatory variables such as policy changes, migration, economic indicators, or population age structure, which may improve the interpretation and forecasting of Singapore's demographic trends.

References

National Population and Talent Division (2025). Population in brief 2025. Government of Singapore.

Singapore Department of Statistics (2023). Births and fertility rates, annual. data.gov.sg. Dataset. Retrieved June 8, 2026, from `https://data.gov.sg/datasets/d_e39eeaeadb571c0d0725ef1eec48d166/view`.

Yamamoto, K., Uchiyama, K., Abe, Y., Takaoka, N., Haruyama, Y., and Kobashi, G. (2025). Birth and pregnancy numbers decreased during the covid-19 pandemic in japan: A time series analysis with the arima model. *Journal of Obstetrics and Gynaecology Research*, 51:e16202.

A Statistical Appendix

A.1 ARIMA Model Form

For a time series X_t, an ARIMA(p, d, q) model can be written using the backshift operator B as

$$\phi(B)(1 - B)^d X_t = \theta(B)Z_t,$$

where Z_t is white noise, $\phi(B)$ is the autoregressive polynomial, and $\theta(B)$ is the moving average polynomial.

A.2 SARIMA Model Form

A SARIMA$(p, d, q)(P, D, Q)_s$ model can be written as

$$\Phi(B^s)\phi(B)(1-B)^d(1-B^s)^D X_t = \Theta(B^s)\theta(B)Z_t,$$

where s is the seasonal or cyclic period.

A.3 Ljung–Box Test

The Ljung–Box test is used to assess whether residual autocorrelations are jointly significant. The null hypothesis is that the residuals are independently distributed. A large p-value indicates that there is no strong evidence of remaining autocorrelation.